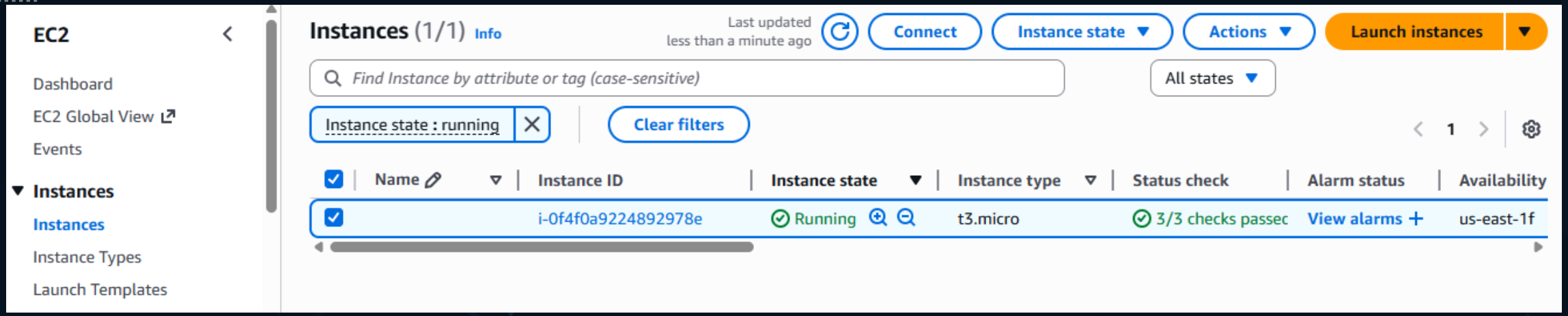
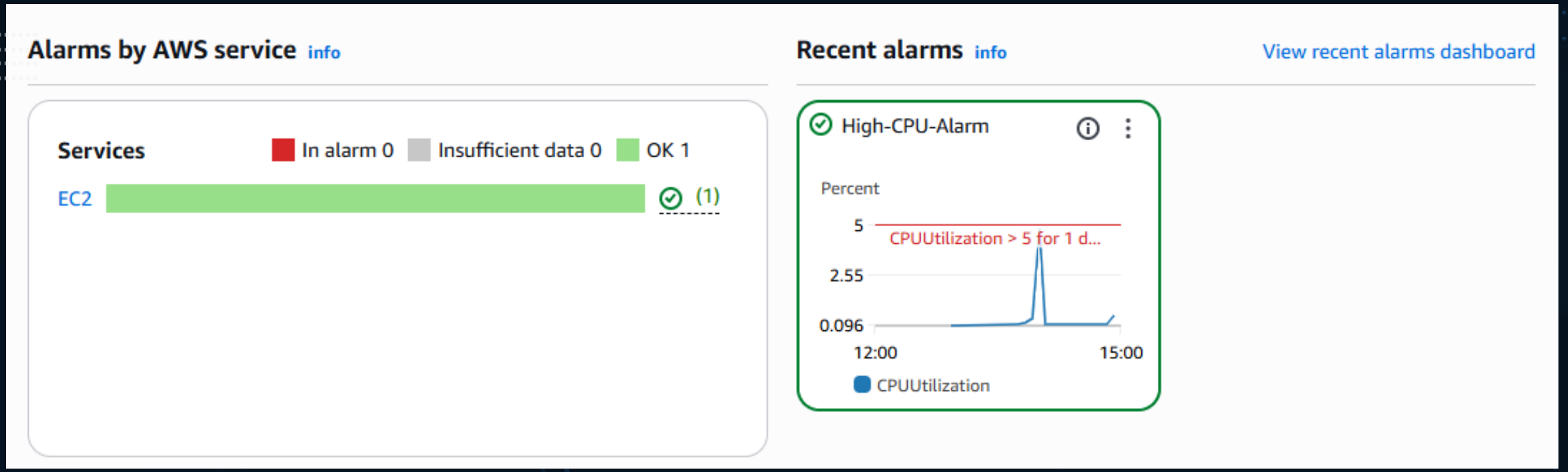
AWS Cloud Monitoring System

MOHAMMED NASSER MOHAMMED NAJI
220218440



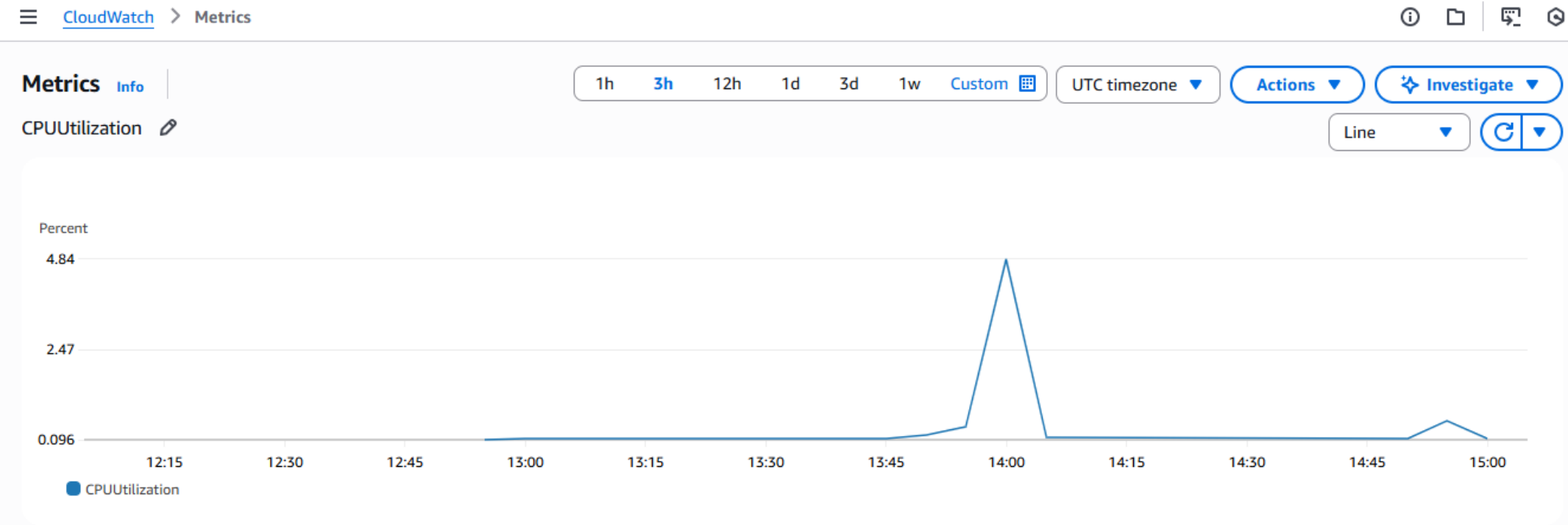
Problem

- Monitoring cloud servers manually is inefficient and unreliable. Without an automated monitoring system, high CPU usage on cloud servers can go unnoticed, which may lead to performance degradation, service downtime, or increased operational costs. Additionally, users may not receive timely notifications when system resources exceed safe limits. Therefore, there is a need for an automated cloud monitoring solution that can track resource usage and send alerts in real .time



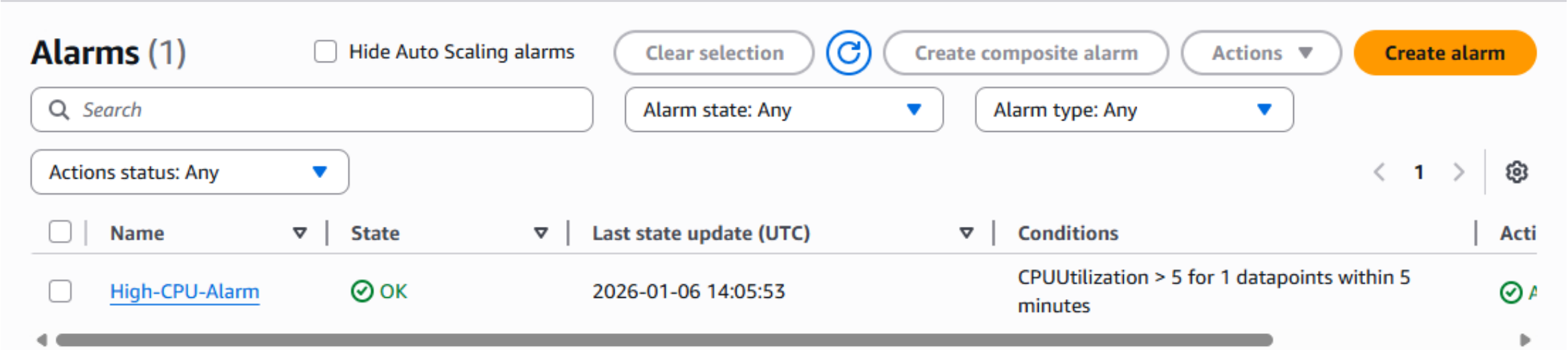
Solution

- The solution is to implement an automated cloud monitoring instance is monitored 2 system using AWS services. An EC using Amazon CloudWatch to track CPU utilization in real time. A CloudWatch alarm is configured to trigger when CPU usage exceeds a predefined threshold. When the alarm is activated, Amazon SNS sends an automatic email notification to alert the user. This solution provides real-time monitoring, automation, and timely alerts, ensuring better performance and system .reliability



Key Features

- Automated CloudWatch alarms for high CPU usage
- CPU utilization2 Real-time monitoring of EC
- Email notifications using Amazon SNS
- Cloud-based solution using AWS services
- Automated alerting without manual intervention
- instance after testing2 Cost-efficient by stopping the EC



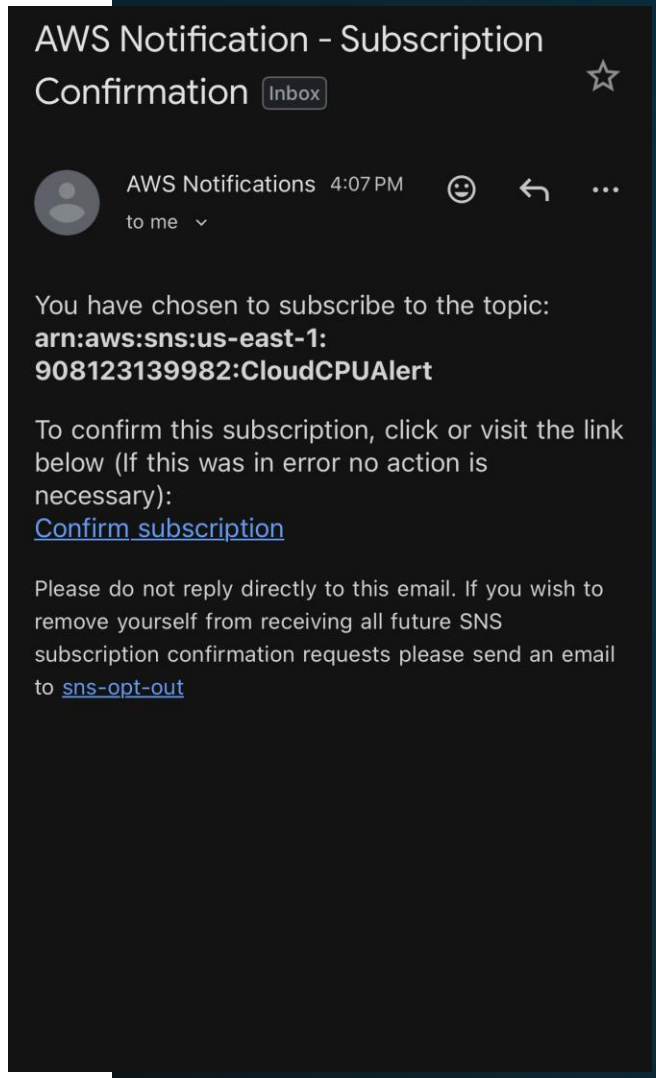
Technical Architecture

The system is built using multiple AWS services working together to provide instance generates 2 automated monitoring and alerting. An Amazon EC performance metrics such as CPU utilization. These metrics are collected by Amazon CloudWatch, which continuously monitors the instance. A CloudWatch alarm evaluates the metrics and triggers an alert when the CPU usage exceeds a predefined threshold. Once triggered, the alarm sends a .notification through Amazon SNS, which delivers an email alert to the user

-

Code Logic (How It Works)

- The system works based on event-driven automation. Amazon CloudWatch continuously monitors the CPU utilization of instance. When the CPU usage 2 the EC exceeds the defined threshold, a CloudWatch .alarm is triggered
- Once the alarm is triggered, an event is generated and sent to AWS services. Amazon SNS receives this event and sends an email notification to the subscribed user. The code logic ensures that alerts are sent automatically .without any manual intervention



Roadmap & Future Improvements

- Monitor additional metrics such as memory usage, disk usage, and network traffic
- Support multiple EC2 instances instead of a single instance
- Add SMS or mobile push notifications in addition to email alerts
- Integrate AWS Auto Scaling to automatically handle high resource usage
- Create a CloudWatch dashboard for better visualization
- Improve alert logic using different thresholds and anomaly detectio

Results of the Project

- The AWS Cloud Monitoring System was implemented and tested instance was 2 successfully. The EC monitored in real time using Amazon CloudWatch, which continuously .tracked CPU utilization
- During testing, high CPU usage was instance. 2 generated on the EC When the CPU utilization exceeded the defined threshold, the CloudWatch alarm changed its state na ,emit emas eht tA .to **ALARM** yllufsseccus saw noitacfiiton liame liame debircsbus eht ot tnes .SNS nozamA gnisu sserdda
- After the CPU usage decreased, the ,etats alarm returned to the **OK** gnirotinom eht taht gnimrfinoc trela htob ni yltcerroc skrow metsys .soiranecs yrevocer dna
- Overall, the project achieved its objectives by providing automated monitoring, real-time alerts, and reliable notification delivery using AWS services